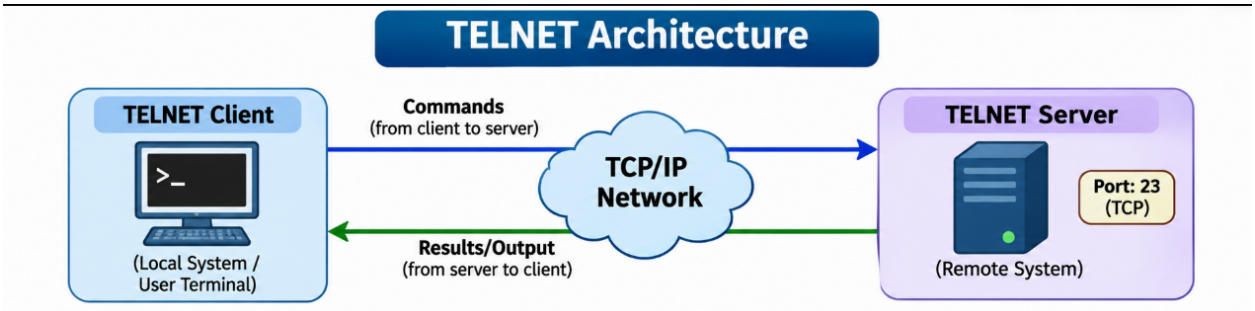
TELNET(TErminaL NETwork)

TELNET (TErminaL NETwork) is an application-layer protocol that allows a user to log in to and access a remote computer over a TCP/IP network. TELNET uses the client-server model. It enables the local computer to communicate with a remote computer as if the user were working directly on the remote system. TELNET does not encrypt data; therefore, SSH is preferred for secure remote login

TELNET consists of three main components:

1. TELNET Client: Runs on the local computer and accepts input from the user.
2. TELNET Server: Runs on the remote computer and processes the commands received from the client.
3. TCP Connection: Establishes a reliable connection between the client and server using TCP port 23.

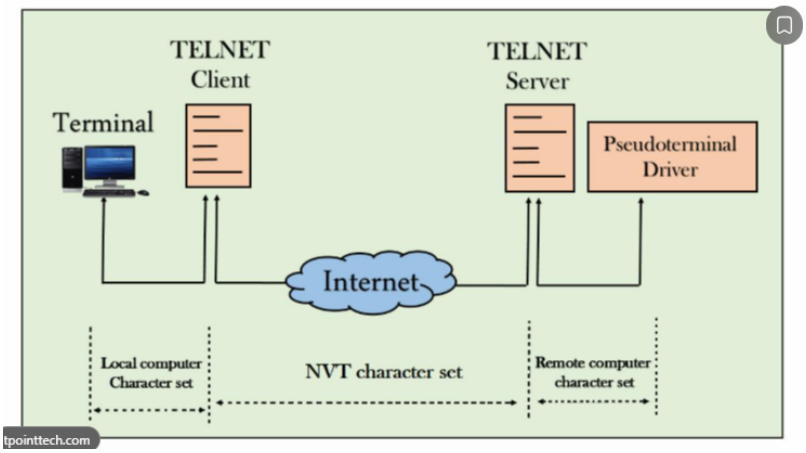


Network Virtual Terminal (NVT)

TELNET uses an intermediate interface called the Network Virtual Terminal (NVT) to enable communication between different types of computers.

- Different computers may use different operating systems, character sets and terminal formats.
- NVT provides a common format for communication between the client and server.
- The client converts its local terminal data into NVT format before transmission.
- The server converts NVT data into the format required by its local system.
- Similarly, the server converts its response into NVT format, and the client converts it into the local terminal format.

Example: A Windows computer can communicate with a Linux server using TELNET because NVT provides a common representation.



TELNET Client-server communication takes place as follows:

1. The user starts the TELNET client on the local computer.
2. The client establishes a TCP connection with the remote TELNET server using port 23.
3. The user enters login credentials (username and password).
4. The client converts the user's input into NVT format and sends it to the server.
5. The server receives the data, converts it into its local format and executes the commands.
6. The server sends the output back to the client through the TCP connection.
7. The client converts the received NVT data into the local terminal format and displays the output to the user.

TELNET also supports option negotiation between the client and server to determine communication parameters.

Examples include:

- Echo : Echo means displaying the characters typed by the user on the terminal screen.
- Suppress Go Ahead : It allows both the client and server to transmit data without waiting for a Go Ahead signal
- Terminal Type : Terminal Type option allows the client to inform the server about the type of terminal being used.
- Window Size : Window Size option communicates the dimensions of the terminal screen.